SALMA BIBI

AI / Machine Learning Engineer | NLP • Generative AI • Agentic AI
Multan, Pakistan | salmamldev56@gmail.com | Open to Remote
LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI / Machine Learning Engineer focused on production systems — not just models. Identifies where AI actually solves the business problem, selects the right approach (ML, deep learning, or GenAI), and owns the full pipeline from data preparation to deployment. Built a full RAG pipeline, agentic AI systems, and deployed ML models via FastAPI and Docker in production.

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript, TypeScript, Dart
- **AI / ML & NLP:** Scikit-learn, XGBoost, LightGBM, TensorFlow, CNN, ResNet50, Transfer Learning, Transformers, BERT, Whisper, Sentence Transformers, Prompt Engineering, RAG Pipelines
- **Generative AI / LLMs:** OpenAI API, Gemini API, Groq (LLaMA), Hugging Face Inference API, LangGraph, Playwright Agent, FAISS Vector Search, Agentic Loops
- **Backend / MLOps:** FastAPI, Node.js, Next.js 14, Docker, Hugging Face Spaces, Streamlit Cloud, MongoDB (ACID), MySQL, PostgreSQL
- **Mobile / Edge:** Flutter, Firebase (Realtime DB, FCM), WebRTC, Android Camera2, ML Kit

PROFESSIONAL EXPERIENCE

AI Engineer | *ClickTake Technologies, Multan, Pakistan* — Apr 2026 – Present

- Built and shipped a GDPR-compliant multi-tenant SaaS repair platform (Next.js 14 + MongoDB) with JWT/RBAC across 4 permission tiers — zero cross-tenant data leakage and zero unauthorized access incidents across all production deployments.
- Improved dashboard query performance by 40% via compound MongoDB indexing, reducing average page load from ~2s to under 0.8s for concurrent enterprise tenants.
- Engineered atomic loyalty-points engine handling 500+ concurrent writes with 100% balance consistency — eliminating ~8 hrs/week of manual PII compliance work via GDPR pipeline.

AI Automation Engineer | *TheEduAssist, Pakistan* — Oct 2025 – Jan 2026

- Designed and deployed NLP-driven mental health kiosk for 200+ users using Transformer-based models — achieved 87% session completion rate, the highest engagement metric at the organization.
- Built custom AI safety layer on Hugging Face Inference API preventing harmful outputs across all user interactions — maintained 100% responsible AI compliance in a sensitive healthcare context.

ML Engineer Intern | *Technocolabs Softwares (Remote)* — Apr 2025 – Jun 2025

- Built ETA prediction system on 50K+ delivery records using XGBoost — reduced prediction error by 23% and improved F1-score from 0.71 to 0.89 through feature engineering on 15+ engineered features.
- Deployed real-time FastAPI inference service achieving sub-200ms latency for 1,000+ daily API calls on Hugging Face Spaces — containerized with Docker for consistent production deployment.

KEY PROJECTS

Multi-Agent Code Review System | *LangGraph Multi-Agent Pipeline* | Link

- **Problem:** Manual code review is slow and inconsistent, and most "AI code review" tools either skip real human oversight or simulate it.
- **Solution:** Built a sequential three-agent pipeline with LangGraph — a Security reviewer flags injection risks and hardcoded secrets, a Logic reviewer checks correctness and edge cases without repeating what Security already found, execution then genuinely pauses using LangGraph's interrupt_before with a checkpointer, and a Test Generator writes pytest tests covering the flagged issues. Ships as a local CLI, a Streamlit UI, and an automatic GitHub Actions review on every pull request.
- **Stack:** Python, LangGraph, Groq LLaMA 3.3 70B, langchain-groq, Streamlit, GitHub Actions

AI Voice Assistant for Document Q&A | *Voice-Driven RAG Pipeline* | Link

- **Problem:** Users needed to ask questions about PDF documents by voice through a backend built to run as a real, multi-user service — not a single-session script.
- **Solution:** Built an async, production-style architecture on FastAPI with Postgres for state and Qdrant for persistent vector search. Redis/Celery run transcription, retrieval, generation, and speech synthesis as background jobs instead of blocking requests. faster-whisper transcribes voice input, Groq/LLaMA 3.3 70B generates context-only answers from retrieved chunks, and Coqui TTS speaks the answer back — with API-key authentication, a Tesseract OCR fallback for scanned PDFs, and Alembic-managed schema migrations.
- **Stack:** Python, FastAPI, PostgreSQL, Qdrant, Redis, Celery, faster-whisper, Groq/LLaMA, Coqui TTS, sentence-transformers, Docker, Alembic

Data Web Scraper Agent — Agentic AI | *Autonomous Browser Automation*

- **Problem:** Manual web scraping requires human decisions at every step and breaks when page structure changes — creating bottlenecks in data collection workflows.
- **Solution:** Built agentic AI system — Gemini reads page HTML, decides next action (click/type/scrape/finish), Playwright executes. Loop continues autonomously until goal complete. Eliminated manual scraping effort entirely for supported sites. Data saved as structured JSON.
- **Stack:** Python, Playwright, Gemini 2.5 Flash, Streamlit | GitHub

PDF Chat Assistant — RAG Pipeline | *From-Scratch RAG* | Link

- **Problem:** Users needed accurate answers from uploaded PDFs without AI hallucinating from outside knowledge — existing tools mixed document content with general AI knowledge.
- **Solution:** Built complete RAG from scratch without frameworks — pdfplumber extraction, 400-word chunking with 50-word overlap, Gemini embeddings, cosine similarity search, Gemini 2.5 Flash for context-only answers. Supports 2 PDFs simultaneously. Reduced answer error rate to 0% from outside hallucination. Deployed live on Streamlit Cloud.
- **Stack:** Python, pdfplumber, Gemini Embedding API, Gemini 2.5 Flash, Streamlit | GitHub

Gadget Buddy AI Chatbot | *Retail Consultation Chatbot* | Link

- AI-powered Streamlit chatbot for gadget consultation, live inventory management, and WhatsApp customer lead capture.
- **Stack:** Python, Streamlit, WhatsApp integration

EDUCATION

B.S. in Software Engineering | Virtual University of Pakistan | CGPA: 3.35/4.00 | Graduated Mar 2025
Final Project (CS619): Tourism & Hotel Booking Management System